ANAGHA PRADEEP

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PROFESSIONAL SUMMARY

Python-focused machine learning and software engineer experienced in meteorological data processing, environmental analytics, sensor-related debugging, and reproducible data pipelines. Integrated IBM PAIRS weather data into crop disease-risk workflows using temperature, humidity, dewpoint, precipitation, wind, and estimated leaf-wetness duration. Built production-oriented Python ML pipelines and real-time embedded automotive software; pursuing an M.S. in Computer Science (Dec 2026).

TECHNICAL SKILLS

Languages: Python, C++, C#, Java, SQL, Bash **Data Analysis:** Pandas, NumPy, DuckDB, MotherDuck, PostgreSQL, ETL
Weather / Scientific Data: IBM PAIRS Weather API, Temperature, Relative Humidity, Dewpoint, Precipitation, Wind, Leaf-Wetness Estimation, Time-Series Features, Data Validation
ML / Statistics: scikit-learn, LightGBM, Random Forest, Feature Engineering, Model Evaluation, Error Analysis, ONNX
Software / Systems: Linux, Git, Docker, AWS S3/Lambda, REST APIs, QNX RTOS, Vector CANalyzer, CANoe, Parasoft, Google Earth Engine

PROFESSIONAL EXPERIENCE

Machine Learning Intern *Mar 2026 – Jul 2026*
Intelinair — Indianapolis, IN

- Integrated weather observations through the **IBM PAIRS Weather API** into Python workflows supporting crop-growth and disease-risk analysis.
- Processed temperature, relative humidity, dewpoint, wind, precipitation, and related weather variables to derive time-dependent environmental features.
- Developed weather-driven disease-risk workflows for Northern Corn Leaf Blight, Gray Leaf Spot, and Southern Rust, including **leaf-wetness duration estimation** from meteorological inputs.
- Built reusable Python pipelines for ingestion, feature computation, model training, evaluation, and batch inference using Pandas, NumPy, DuckDB, and MotherDuck.
- Engineered 41 agronomic and weather features for LightGBM growth-stage classifiers, achieving **76.5% exact / 95.1% within-one-stage** accuracy for corn and **75.9% / 92.5%** for soybean.
- Developed features including cumulative growing degree days, photoperiod, humidity, dewpoint, and precipitation; investigated feature importance and prediction errors.
- Applied field-grouped validation and future-year holdouts to evaluate generalization and reduce leakage between training and evaluation data.
- Automated experimental runs, feature computation, performance reporting, and prediction generation using reusable Python modules and command-line tools.
- Developed gradient-boosting planting-date regression workflows using thermal-time and temporal features; exported models to ONNX for portable inference.

Research Assistant — Geospatial Machine Learning *Jan 2026 – Present*
Indiana University Indianapolis

- Develop Python and Google Earth Engine workflows to process Sentinel-1 radar and Sentinel-2 optical measurements for surface-water classification using Random Forest models.
- Extract spectral indices (NDWI, MNDWI, NDVI), radar backscatter, and terrain features to characterize environmental conditions across satellite observations.
- Integrate JRC Global Surface Water and Dynamic World reference products; investigate reference-label disagreements and uncertainty in environmental classification.
- Design geographic block-based validation splits to limit spatial leakage; compare sensor feature groups using precision, recall, F1, and error analysis.
- Process large GeoTIFF and satellite raster datasets to investigate permanent and seasonal surface-water variation across Indiana.

Software Engineer *Jul 2023 – Dec 2024*
Stellantis — Bangalore, India

- Developed C++, C#, Qt, and Unity applications for embedded automotive digital instrument clusters, integrating real-time vehicle measurements with dashboard interfaces.
- Diagnosed CAN-bus signal and communication issues using Vector CANalyzer and CANoe, investigating signal behavior and application-level defects.
- Debugged, integrated, tested, and validated applications on QNX RTOS, optimizing execution efficiency, resource utilization, and responsiveness under real-time constraints.
- Designed and maintained software components connecting vehicle signals to dashboard applications with an emphasis on reliability and fault investigation.
- Performed static analysis and software verification with Parasoft; collaborated with integration and testing teams to resolve defects and support releases.

Graduate Research Assistant — Geospatial Analytics *Aug 2025 – Dec 2025*
Indiana University Indianapolis

- Automated multiyear agricultural land-cover analysis in Google Earth Engine using MODIS, USDA Cropland Data Layer, and Census TIGER boundaries.
- Applied raster masking, clipping, pixel-level comparisons, and area statistics to evaluate year-over-year changes in environmental datasets.
- Built reproducible raster-processing and visualization workflows for large-scale geographic observations.

Machine Learning Intern *Dec 2022 – Mar 2023*
HighRadius — Chennai, India

- Developed Python ML workflows for payment-date prediction using historical invoice and transaction data, including preprocessing, feature engineering, and evaluation.
- Transformed invoice attributes and payment histories into model-ready features for predictive analytics.

SELECTED PROJECTS

Weather-Based Crop Disease Risk Modeling | Python / IBM PAIRS *2026*

- Developed meteorological feature-processing workflows using IBM PAIRS weather data to support crop disease-risk estimation.
- Estimated leaf-wetness duration from weather variables and examined the distinction between observed atmospheric conditions and inferred leaf-surface moisture.
- Incorporated temperature, humidity, dewpoint, wind, and precipitation into disease-risk logic informed by agronomic literature.

Surface Water Detection | Multisensor Environmental Data *2026*

- Combined Sentinel-1 radar and Sentinel-2 multispectral inputs with reference water labels for Random Forest classification experiments.
- Evaluated optical-only, radar-only, and combined feature sets with spatially separated validation to investigate sensor contributions and error patterns.

EDUCATION

Master of Science, Computer Science *Jan 2025 – Dec 2026 (expected)*
Indiana University Indianapolis GPA: 3.5/4.0

Bachelor of Technology, Computer Science and Engineering (Big Data Analytics) *May 2019 – May 2023*
SRM Institute of Science and Technology CGPA: 9.2/10.0